Practicum Sprint 2 - Project Planning

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***Abstract*—**The main objective of the project is to create a dashboard that monitors Type 1 Diabetes as a patient.This report includes a one page white paper discussing the project topic, technical design, and implementation plan.

1 PART 1 - PROJECT TOPIC

Type 1 diabetes is a chronic condition (Mayo Clinic, 2024). Pancreas with this condition makes little or no insulin. Insulin is a hormone used to allow glucose to enter cells to produce energy. Currently, there's no cure for type 1 diabetes. Therefore, patients with type 1 diabetes continuously need to manage Blood Sugar Level by using insulin, controlling diet and lifestyle. It is critical to use all scheduled insulin for type 1 diabetes patients. High or low blood sugar levels can lead to severe hyperglycemia or diabetic ketoacidosis.

This project will create a dashboard that is monitoring Type 1 Diabetes. It will include quick access for following items:

- Recent glucose levels
- Insulin Prescription
- Long-term glucose control
- Diabetes care plan

Type 1 diabetes patients always need to monitor glucose levels and consider how many insulins are left because one missed insulin is directly critical to their life. They also need to plan their long-term glucose control and diabetes care plan to minimize irregular glucose spikes during their daily activities.

Dashboard could include 5 screens: Main, Recent Glucose Levels, Insulin Prescription, Long-term Glucose Control, Diabetes Care Plan.

Main screen includes possibly 4 sections to display key factors for each section. For example, section 1 displays recent glucose levels with recorded time and indicates the difference from previous recorded glucose levels, color option would be great to visualize its importance. Section 2 displays recent insulin prescription with date. For main screen purpose, it is enough to display last 2 prescriptions. Section 3

displays how patient's glucose levels are changed over time. It can be a bar type or trend graph with daily/weekly/monthly option. Section 4 displays diabetes care plan. One of ideal information for this section is displaying a time indicator if patient needs to get insulin within certain time periods. On left side of main screen, it will have 5 icons that direct to each detailed menu.

Additionally, before accessing main screen, login screen is also required. This screen requires ID and password. If a patient doesn't have it, they need to create a new ID, including name, birth date, ID.

2 PART 2 - TECHNICAL DESIGN

2.1 Tools/Technology

The application will be a full-stack web application which consists of frontend, backend, and database. For frontend, language will be TypeScript with React framework. For backend, language will be Python with either Flask or Django framework, depending on the project complexity as it grows. There is a possibility to switch to Java/Spring depending on the project needs. For database, PostgreSQL will be used to leverage the existing table setups from previous labs. There is a possibility of utilizing MongoDB if it is much easier to document-store the FHIR datasets. Deployment will be done through Docker containerization.

2.2 Datasets and Data Sources

The application will leverage FHIR dataset with following resources:

- Patient: name, gender, birth date, patient ID, age, etc.
- Condition: diagnosis, status, etc.
- Observation: status, values, unit, effectiveDateTime, etc.
- MedicationRequest: dosageInstruction, dosageQuantity, etc.
- CarePlan: description, code, activity, etc.

2.3 Architecture Diagram

Below figure 1. is the architecture diagram for the application.

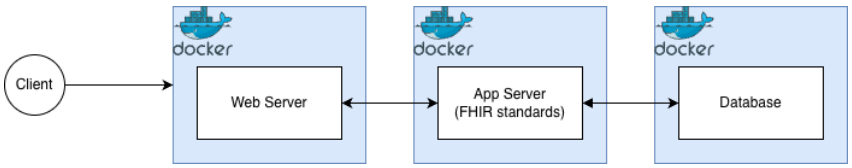

Figure 1—High level architecture diagram for the Type 1 Diabetes web application.

2.4 Screen Mock-up

Figure 2. and figure 3. are the mock screens for the application.

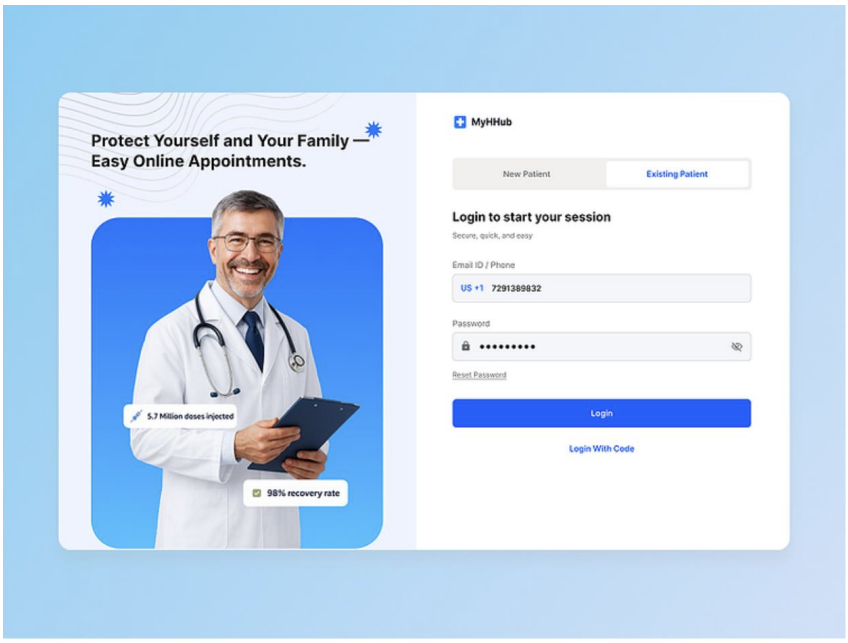

Figure 2—Login screen that the patient will face before accessing the dashboard.

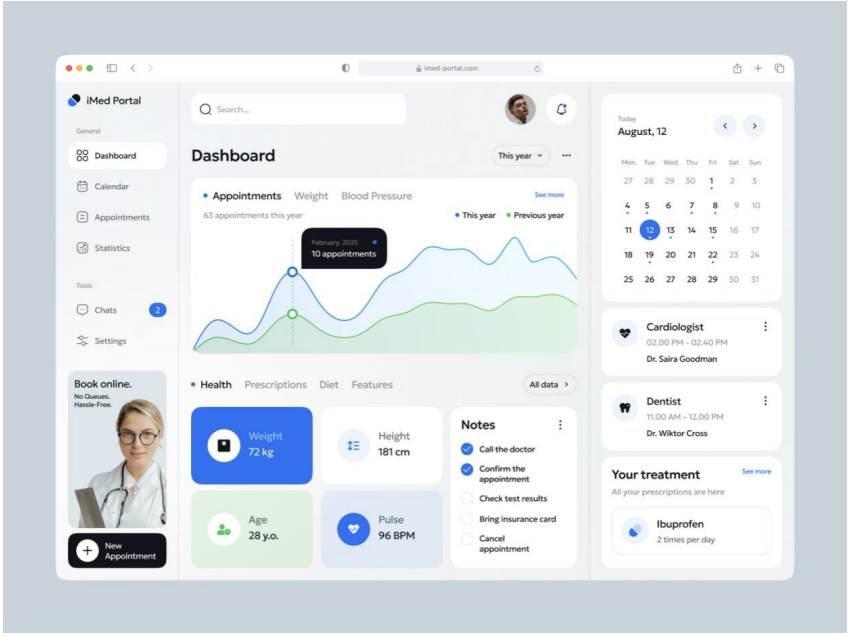

Figure 3—Dashboard that captures details such as appointments, prescriptions, and primarily details related to Type 1 Diabetes.

3 PART 3 - IMPLEMENTATION PLAN

3.1 Project Tasks

Below is the link to the project tasks:
Type 1 Diabetes Monitor Dashboard
Figure 4. below is a screenshot of the tasks in case the link is broken.

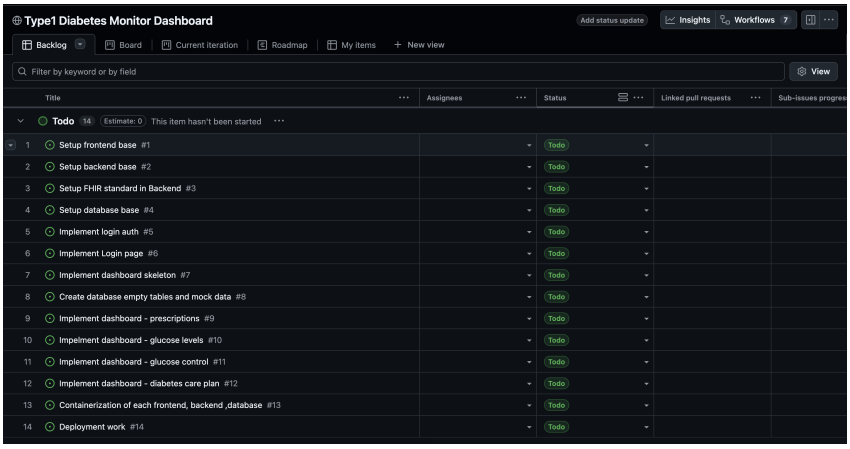

Figure 4—Screenshot of project tasks created on Github Projects.

3.2 Project Timeline

Figure 5. below displays a visualized project timeline with different phases.

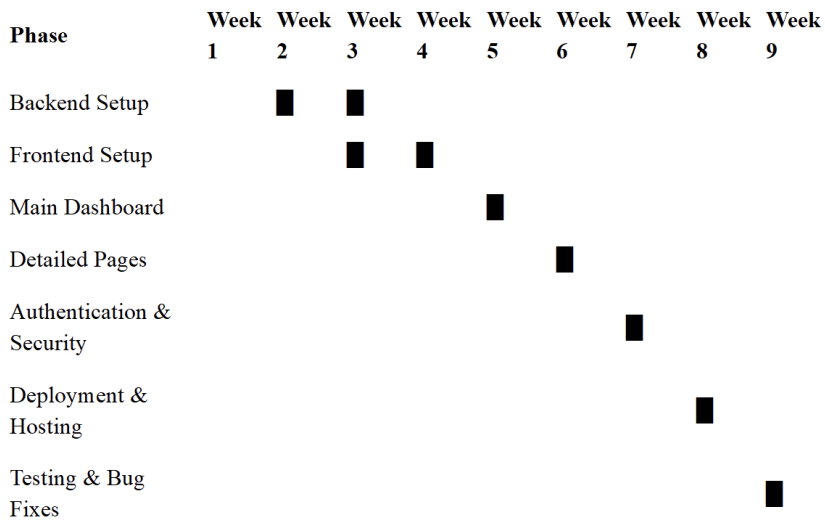

Figure 5—Project timeline visualization. Submission of this project is week 2, which iterates until week 9, the final submission date.

3.3 Needs/Risks

Key risk in this project is both developers of the project are not very familiar with FHIR standards and resources, which could impact the technical stack choice of the project, which may result in timeline impact. Developers need to be regularly investing time on learning unfamiliar concepts to mitigate the risk.

4 REFERENCE

Type 1 diabetes - Symptoms and causes. (n.d.). Mayo Clinic. https://www.mayoclinic.org/diseases-conditions/type-1-diabetes/symptoms-causes/syc-20353011